PaaS platform automatically can add more servers and sync load balancers to maintain the optimal efficiency of your running application. Once the load or requests on your application reduce, the additional servers are de-provisioned automatically.

Easy upgrades: A typical application deployment may rely on a number of components that individually need to be tracked, maintained, upgraded and re-integrated over time. The PaaS provider abstracts the management, timely security and patch upgrades of the PaaS platform so that the users do not waste time and incur risks by doing these tasks themselves.

A vast ecosystem: A PaaS platform provides a rich set of tools, frameworks and UIs for managing and building your applications on the cloud. A developer has the freedom to choose from a variety of development languages such as Java, .NET, Node.js, PHP, Python, Ruby, etc, all pre-installed, pre-configured and ready for use.

No vendor lock-in: A PaaS platform provides the freedom of choice to developers to move to different PaaS platforms without being locked in to any vendor.

PaaS providers

There are a variety of PaaS providers in the market today, each providing their own set of tools and frameworks to develop applications. Here's a look at some of the market leaders in this space.

Windows Azure: Windows Azure is a PaaS platform developed by Microsoft to directly compete with other IaaS and PaaS providers such as Amazon Web Services (AWS) and Google App Engine (GAE). Windows Azure provides a rich set of tools and frameworks, which are specific to both Microsoft and third parties, that can help developers build their applications on the cloud.

Additionally, it also provides media encoding services, multi-tier applications, and automated deployment services using Git and Eclipse. Windows Azure also provides SQL Database as a service in another offering called SQL Azure Database. This can help developers integrate their applications with a scalable database that supports Microsoft's Active Directory components, Microsoft System Center, Hadoop Processing Framework, etc.

Azure provides the Blob store and table-like structures to store your application data. It has a REST, XML and http-based API that can be used by developers to extend their application's functionality with the Azure framework.

The pricing is simple and also offers billing on a pay-as-you-go model as well as pre-paid plans.

Amazon Elastic Beanstalk: Amazon Elastic Beanstalk is a PaaS-like service from Amazon Web Services (AWS), a major IaaS provider. Beanstalk provides users the ability to quickly deploy their applications and manage them on the AWS Cloud. It provides an easy to use Web UI with which developers can upload their code as a WAR file, tune in a couple of necessary parameters, and finally launch their application on AWS. Beanstalk will automatically provision the virtual machine instances, monitor the workloads of your application and balance load across the instances effectively.

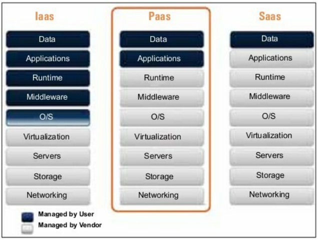

Figure 1: PaaS stack

Figure 2: Elastic Beanstalk management console

Beanstalk provides users complete control over their AWS resources that power the application. Users have the ability to browse through the application log files, monitor the application health and even pass the necessary run time values to their applications.

Users also have the freedom to directly publish their code on Elastic Beanstalk using tools such as Visual Studio, Eclipse IDE and Git.

Since AWS offers loosely coupled services, Elastic Beanstalk can be integrated with other AWS services such as Simple Notification Service (Amazon SNS), Simple Storage Service (Amazon S3), Auto-Scaling, Elastic Load Balancer (ELB), etc.

Elastic Beanstalk currently can deploy applications written in .NET, Java, Node.js, Ruby, Python and PHP.

There is no additional charge for using Amazon Elastic Beanstalk. Users just have to pay for the underlying AWS resources that they consume, such as EC2, S3, ELB, etc.

Google App Engine: Google App Engine (GAE) is a PaaS offering that enables users to host their Web applications on Google's infrastructure, i.e., Google's globally present data centres in a 'Sandbox' like environment. GAE offers auto-scaling of resources, and load balancing that enables your Web